

Senior Thesis Proposal: Examining Tidal Deformability of Neutron Stars by Simulating Inspiral Degeneracy of Differing Internal Structures

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The inspiral of a binary neutron star system creates a gravitational wave signal that can be observed by gravitational wave observatories like LIGO and the upcoming Cosmic Explorer. This wave signal is currently characterized by the singular parameter λ , or the tidal deformability. This factor measures the quadrupole deformation of the star as it responds to a perturbing tidal field of its partner in the inspiral. The question to be examined by my thesis work is whether neutron star gravitational wave signals are degenerate for systems that have identical λ . Specifically, we will model stars that have the same λ , i.e. the same macrophysics, but exhibit different internal behaviors: different internal energy densities (ϵ), different equations of state (EOS), and different compactnesses ($c = \frac{m}{r}$), all of which are factors that describe the microphysics of the system.

The question of degenerate waveforms for identical tidal deformabilities has great implications on the information we can extract from gravitational wave observations. If waveforms are indeed degenerate, or rather, if the tidal deformability is indeed the single parameter that characterizes the wave signal, then observation of neutron star binary inspirals will not convey any information about the internal structure of the star. If the waveforms are not degenerate, then the waveform may indicate the internal structure of the star via a parameter other than the tidal deformability. For my senior thesis, I will be investigating this question through use of SpECTRE's Tolman–Oppenheimer–Volkoff (TOV) equation solver, and I will develop code that compares the expected waveforms stars for a given λ .